

Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - II

Course Code : CSE 0533 1201 (CSE) & PHY 0533 1201 (EEE)

Term: Spring – 2024	Date: 01.01.2024	Time: 03.30 PM	Room: 327	Activity: Lecture 1
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Topics to be discussed: Principle of temperature measurements, Platinum resistance thermometer

Target CLO & PLO : CLO1,2,4 & PLO1,2,4

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define Heat & Temperature
- ❖ Explain the principle of temperature measurement
- ❖ Define Thermometric properties & substance
- ❖ Define resistance thermometer & how it works?
- ❖ Solve assigned recommended mathematical problems

Methods of Teaching:

Mode/Activities	Time
Introductory Speech	15 min
Delivery of the lesson plan & OBE course outline	5 min
Interactive discussion	10 min
Lecture Presentation on Topics	30 min
Group activity	15 min
Feedback session	10 min
Others = 5 min	5min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

- 1) The most usable substance for the resistance thermometer is
a) Iron b) Cobalt c) Carbon d) Platinum e) nickel
- 2) Which is the thermometric property?
a) Molecular weight b) Intermolecular force c) Binding energy d) Enthalpy e) Volume

Sample Questions (SAQ):

- 1) Define Heat & Temperature
- 2) State the principle of temperature measurement.
- 3) What are the thermometric properties? Give example.
- 4) What is the thermometric substance? Give example.
- 5) Define resistance thermometer.
- 6) Give properties of platinum resistance thermometer.
- 7) Establish the relation - $\theta = \frac{R_{\theta} - R_0}{R_{100} - R_0} \times 100$
- 8) The resistances of a resistance thermometer at 0°C and 100°C are 8Ω and 20Ω respectively. If the thermometer is placed in furnace, it shows 32Ω resistance. Find the temperature of furnace.
- 9) A platinum resistance thermometer reads 2Ω and 2.73Ω at ice point and steam point respectively. Find the temperature at which it reads 4.83Ω.

Reference Books: 1. Giasuddin ahmed – Physics for Engineers (Part-1)
2. Halliday Resnick – Fundamental of Physics, 2nd edition

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 01-64
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 01-36